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Gastrointestinal Graft-versus-Host Disease Is a Risk Factor for Postengraftment Bloodstream Infection in Allogeneic Hematopoietic Stem Cell Transplant Recipients



Yasuo Mori¹, Goichi Yoshimoto¹, Ruriko Nishida¹, Takeshi Sugio¹, Kohta Miyawaki¹, Takahiro Shima¹, Yoji Nagasaki¹, Noriko Miyake¹, Yukiko Harada¹, Yuya Kunisaki², Kenjiro Kamezaki², Akihiko Numata¹, Koji Kato¹, Motoaki Shiratsuchi³, Takahiro Maeda², Katsuto Takenaka¹, Hiromi Iwasaki¹, Nobuyuki Shimono¹, Koichi Akashi^{1,2}, Toshihiro Miyamoto^{1*}

¹ Medicine and Biosystemic Science, Kyushu University Graduate School of Medical Sciences, Fukuoka, Japan

² Center for Cellular and Molecular Medicine, Kyushu University Hospital, Fukuoka, Japan

³ Medicine and Bioregulatory Science, Kyushu University Graduate School of Medical Sciences, Fukuoka, Japan

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ABSTRACT

Bloodstream infection (BSI) is a well-known cause of morbidity and mortality in allogeneic hematopoietic stem cell transplant (allo-HSCT) recipients. Here, we conducted a retrospective study to assess the morbidity, etiology, risk factors, and outcomes of BSI in the postengraftment period (PE-BSI) after allo-HSCT. Forty-three of 316 patients (13.6%) developed 57 PE-BSI episodes, in which 62 pathogens were isolated: Gram-positive bacteria, gram-negative bacteria, and fungi, respectively, accounted for 54.8%, 35.5%, and 9.7% of the isolates. Multivariate analysis revealed methylprednisolone use for graft-versus-host disease (GVHD) prophylaxis (odds ratio [OR], 6.49; 95% confidence interval [CI], 1.49 to 28.2; $P = .013$) and acute gastrointestinal GVHD (GI-GVHD) (OR, 8.82; 95% CI, 3.99 to 19.5; $P < .0001$) as risk factors for developing PE-BSI. This finding suggested that GI-GVHD increases the risk of bacterial translocation and subsequent septicemia. Moreover, among patients with GI-GVHD, insufficient response to corticosteroids, presumably related to an intestinal dysbiosis, significantly correlated with this complication. Patients with PE-BSI presented worse outcome compared with those without (3-year overall survival, 47.0% versus 18.6%; $P < .001$). Close microbiologic monitoring for BSIs and minimizing intestinal dysbiosis may be crucial to break the vicious cycle between GI-GVHD and bacteremia and to improve transplant outcomes especially in patients who require additional immunosuppressants.

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INTRODUCTION

Bloodstream infection (BSI) is a common complication in patients after allogeneic hematopoietic stem cell transplantation (allo-HSCT) and can be fatal without appropriate and expedient treatment, especially in the pre-engraftment period [1–3]. Intensive chemotherapy and irradiation allows bacteria to enter the systemic circulation via disrupted mucosal barriers of the gastrointestinal (GI) tract and/or oral cavity, providing an inflammatory cytokine milieu fit for T-cell proliferation and activation. This culminates in the development of acute graft-versus-host disease (GVHD) after the engraftment of donor cells [4]. Acute GVHD targets tissue stem cells [5], which could

delay repair of damaged mucosa and require additional intense immunosuppressive therapy. Thus acute GVHD is considered a risk factor for BSI during the postengraftment phase (PE-BSI) [6–11]. A pediatric study identified a direct relationship between GI-GVHD and PE-BSI [12]. However, other studies could not corroborate this association [13–15].

Recently, allo-HSCT has been applied to patients with older age, impaired organ function, or prior history of allo-HSCT because of improvement of supportive care and reduced-intensity conditioning (RIC) regimens with tolerable toxicity and an immune-mediated graft-versus-leukemia effect. In addition, allo-HSCT from alternative donors who share a single HLA haplotype has increased because of the use post-transplant cyclophosphamide [16,17] or antithymocyte globulin (ATG) [18,19]. Although this allows for “high-risk” patients to receive allo-HSCT, there is still an increased risk of severe opportunistic infections and acute GVHD. In our retrospective analysis we evaluated the incidence, clinical characteristics,

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* Correspondence and reprint requests: Toshihiro Miyamoto, MD, PhD, Medicine and Biosystemic Science, Kyushu University Graduate School of Medical Sciences, Fukuoka, Japan 3-1-1 Maidashi, Higashi-ku, Fukuoka, 812-8582, Japan.

E-mail address: toshmiya@intmed1.med.kyushu-u.ac.jp (T. Miyamoto).

risk factors, and outcomes of PE-BSI patients in a more recent cohort of allo-HSCT recipients.

PATIENTS AND METHODS

Patients

We reviewed the medical records of 346 patients who underwent allo-HSCT at Kyushu University Hospital between July 2009 and December 2016. Patients who died before neutrophil engraftment, defined as an absolute neutrophil count $>0.5 \times 10^9/L$ for at least 3 consecutive days after transplantation [6–9,13,15–17], or experienced graft failure, defined as no engraftment on day 28 after transplantation, were excluded ($n = 30$). Patient characteristics included this analysis ($n = 316$) are listed in Table 1. Underlying diseases were myelodysplastic syndrome/acute myelogenous leukemia ($n = 144$); myeloproliferative neoplasms, including chronic myelogenous leukemia ($n = 13$); acute lymphoblastic leukemia ($n = 52$); malignant lymphoma ($n = 84$); plasma cell disorders ($n = 7$); aplastic anemia ($n = 6$); and others ($n = 10$). In total, 128 patients with acute leukemia (acute myelogenous leukemia and acute lymphoblastic leukemia) in remission, myelodysplastic syndrome without excess of blasts, myeloproliferative neoplasm and chronic myelogenous leukemia in the nonleukemic phase, malignant lymphoma in the first or second at least partial remission, and plasma cell disorders in at least partial remission were considered to be standard risk. All others ($n = 188$), including patients with a history of allo-HSCT ($n = 78$), were considered high risk. This study was approved by the institutional review board of Kyushu University Hospital (approval no. 29-179).

Treatment Procedures

A total of 133 patients received myeloablative conditioning regimens that contained 12 Gy total body irradiation or 12.8–mg/kg busulfan. The remaining 183 patients received a fludarabine-based RIC regimen (Table 1). Low-dose total body irradiation (2–4 Gy) was added to conditioning regimens for 136 patients. Stem cell sources included granulocyte colony-stimulating factor–mobilized peripheral blood ($n = 129$), bone marrow ($n = 126$), or cord blood ($n = 61$). Stem cells were obtained from HLA fully matched related donors ($n = 64$); partially matched related donors ($n = 68$), including 54 haploidentical donors; fully matched unrelated donors ($n = 77$); or mismatched unrelated donors ($n = 107$).

A total of 101 recipients had received 1 or more previous HSCTs as listed in Table 1. The sources of stem cells for prior transplants were 23 autologous and 78 allogeneic. GVHD prophylaxis was as follows: 195 patients received calcineurin inhibitor plus short-term methotrexate (10–7–7–7 mg/m² on days 1, 3, 6, and 11 [optional], respectively), 71 patients received calcineurin inhibitor plus mycophenolate mofetil, and 50 patients received calcineurin inhibitor plus methylprednisolone (mPSL). ATG ($n = 51$) or post-transplant cyclophosphamide ($n = 13$) was added mainly in recipients transplanted from mismatched donor. Granulocyte colony-stimulating factor was given from day +1 (cord blood) or day +5 (other sources) and continued until the absolute neutrophil count was $>1.0 \times 10^9/L$ for at least 3 consecutive days.

Prophylactic Use of Antimicrobial Agents

Patients were treated in a transplantation unit, where high-efficiency particulate air filters were appropriately equipped,

Table 1
Pretransplantation Characteristics of the 316 Patients

	Total (N = 316)	BSI		P
		Yes (n = 43)	No (n = 273)	
Median age, yr (range)	49 (16–72)	56 (22–69)	48 (16–72)	.213
Sex, male/female	169/147	27/16	142/131	.188
Underlying disease				.87
MDS/AML	144	20	124	
CML/MPN	13	0	13	
ALL	52	7	45	
Malignant lymphoma	84	14	70	
Plasma cell disorders	7	1	6	
AA	6	0	6	
Others	10	1	9	
Risk status at transplantation				.035
Standard/high	127/189	11/32	116/157	
Conditioning regimen				.018
Myeloablative/RIC	133/183	11/32	122/151	
Graft source				.069
BM/PB/CB	126/129/61	15/23/5	111/106/56	
Type of donor				.167
Matched related/partially matched related (haplo)	64/68 (54)	6/16 (12)	58/52 (42)	
Matched unrelated/mismatched unrelated	77/107	9/12	68/95	
Prior transplantation				.252
No	215	26	189	
Autologous/allogeneic	23/78	4/13	19/65	
GVHD prophylaxis				
CsA/Tac	70/246	5/38	65/208	.074
sMTX, yes/no	195/121	20/23	175/98	.059
MMF, yes/no	71/245	7/36	64/209	.295
mPSL, yes/no	50/266	16/27	34/239	<.0001
ATG, yes/no	51/265	12/31	39/234	.024
PTCy, yes/no	13/303	0/43	13/260	.143

MDS/AML indicates myelodysplastic syndrome/acute myelogenous leukemia; CML/MPN, chronic myelogenous leukemia/myeloproliferative neoplasm; AA, aplastic anemia; BM/PB/CB, bone marrow/peripheral blood/cord blood; CsA/Tac, cyclosporine A/tacrolimus; sMTX, short-term methotrexate; MMF, mycophenolate mofetil; PTCy, post-transplant cyclophosphamide.

at least until engraftment. As bacterial prophylaxis, oral levofloxacin (500 mg/day) was generally administered from the beginning of the conditioning regimen to engraftment. Oral fluconazole or intravenous echinocandins (micafungin or caspofungin) and oral acyclovir were routinely given as prophylaxis for fungal infections and herpes simplex virus reactivation, respectively. Oral trimethoprim/sulfamethoxazole or inhalation of pentamidine was prophylactically used for *Pneumocystis jirovecii* infection from the time of engraftment until the cessation of immunosuppressants.

Blood Cultures and BSI Diagnosis

For all patients, surveillance culture using throat and rectal swabs were routinely performed once a week. Urine or sputum samples were collected and tested if patients had genitourinary or respiratory tract infection symptoms. In addition, when a febrile episode (axillary temperature $\geq 37.5^{\circ}\text{C}$) occurred, 2 sets of blood cultures were obtained from patients' peripheral vein and central vein catheter. It was repeated every 3–days if fever persisted. For patients who received corticosteroids, supplemental blood cultures were added to regularly scheduled blood cultures based on the physicians' decision.

As previously reported [15], BSIs were defined as the isolation of a bacterium or fungus from at least 1 blood culture. For common skin contaminants, such as coagulase-negative staphylococci or *Corynebacterium*, 2 separate positive blood cultures were required for BSI diagnosis. We basically removed the central venous catheter when the blood culture test results became positive. In this study we defined PE-BSI as an infection that occurred from neutrophil engraftment until day 180. Regarding patients who developed multiple PE-BSI episodes, clinical data of the first episode were used in this analysis.

Acute GVHD

Acute GVHD was diagnosed and graded according to well-established criteria [20]. Treatments against acute GVHD were decided at the physicians' discretion; in general, grade II GVHD without lower GI involvement or bilirubin elevation was treated with up to 1–mg/kg mPSL. However, 2–mg/kg mPSL was used for grade II GVHD with lower GI involvement or grade III/IV GVHD. Treatment response was assessed on days 5 to 7 after the initiation of mPSL treatment [21]. When GVHD manifestations improved, steroid doses were gradually tapered down, but if GVHD manifestations did not improve, second-line treatments (eg, etanercept, mycophenolate mofetil, and ATG) were considered.

Statistical Analysis

For univariate analysis, we examined categorical variables, including age, sex, underlying diseases, disease status at allo-HSCT, conditioning regimen, graft source, donor type, prior HSCT, GVHD prophylaxis, severity and involved organs in acute GVHD, treatment of acute GVHD, and response to initial GVHD treatment, using the chi-square test. Factors with a significant association with PE-BSI were subjected to a multivariate analysis using the Cox proportional hazards model. Overall survival (OS) after allo-HSCT was measured from the date of stem cell infusion until the date of death or the date of the last follow-up at which the patient was known to be alive. The survival period was calculated using the Kaplan-Meier method. $P < .05$ was considered to be statistically significant. All statistical analyses were performed using EZR (Saitama Medical Center, Jichi Medical University), a graphic user interface for R (version

3.2.4; The R Foundation for Statistical Computing, Vienna, Austria) [22].

RESULTS

PE-BSI Incidence

In our study 43 of 316 allo-HSCT recipients (13.6%) developed PE-BSI, of which 30 patients had 1 PE-BSI episode, 12 patients had 2 episodes, and 1 patient had 3 episodes. PE-BSI with multiple microbial detection occurred in 4 cases. Twelve of 43 PE-BSI episodes (27.9%) were identified at surveillance blood cultures. Five cases had low-grade fever (37.0 to 37.5°C), whereas 7 cases were afebrile.

Isolated Pathogens and Antimicrobial Susceptibilities

Among 62 isolates implicated in PE-BSIs, 34 (54.8%) were gram-positive bacteria, 22 (35.5%) were gram-negative bacteria, and 6 (9.7%) were fungi (Table 2). *Staphylococcus epidermidis/aureus* and *Enterococcus faecium* were frequently isolated in gram-positive bacteria, whereas the most common isolate in gram-negative bacteria was *Klebsiella pneumoniae* followed by *Stenotrophomonas maltophilia*. These causative microorganisms had been detected in 20 prior surveillance culture samples collected from 18 patients (41.9%): 3 sputum, 5 throat swab, 3 rectal swab, 7 urine, and 2 other samples.

Eighteen patients developed PE-BSI at a median of 25–days (range, 2 to 65) after the discontinuation of antibiotics. Twenty-one of the remaining 25 patients developed PE-BSIs because of a treatment with ineffective antimicrobials, whereas 4 patients received antibiotics to which, at least in *in vitro* assay, isolates were susceptible. All fungi isolated were non-*Candida albicans* spp. and occurred despite prophylactic echinocandins administration, indicating that these episodes were breakthrough infections (Table 2).

Pretransplantation Characteristics of PE-BSI Patients

As shown in Table 1, GVHD prevention strategies were strongly associated with PE-BSI incidence; to note, 16 of 50 patients (32.0%) who received mPSL developed PE-BSI, which was significantly more frequent than in the 27 of 266 patients (10.2%) who did not receive mPSL ($P < .0001$). Twelve of 51 patients (23.5%) treated with ATG and 31 of 265 patients (11.7%) not treated with ATG developed PE-BSI, respectively ($P = .024$). Tacrolimus was more frequently used than

Table 2

Microorganisms Isolated from Bloodstream Among Patients with Acute GVHD

Organism	n	Organism	n
Gram-positive bacteria	34	Gram-negative bacteria	22
<i>Staphylococcus epidermidis</i>	18	<i>Klebsiella pneumoniae</i>	8
<i>aureus</i>	8	<i>Stenotrophomonas maltophilia</i>	5
<i>caprae</i>	6	<i>Pseudomonas aeruginosa</i>	3
<i>capitis</i>	3	<i>Enterobacter cloacae</i>	2
<i>Enterococcus faecium</i>	1	<i>Bacteroides species</i>	2
<i>faecalis</i>	8	Others	2
<i>raffinosis</i>	5		
<i>Corynebacterium striatum</i>	2	Fungi	
<i>jeikeium</i>	1	<i>Candida</i>	6
Unknown	4	<i>parapsilosis</i>	3
<i>Mycobacterium species</i>	2	<i>guilliermondii</i>	2
Others	2	<i>rugosa</i>	1
		Total	62

Table 3
Post-Transplantation Characteristics of the 316 Patients

	BSI		P
	Yes (n = 43)	No (n = 273)	
Median onset of BSI, day (range)	65 (18–154)	NA	NC
Median ANC, $\times 10^9/L$ (range)	2.34 (0.14–12.48)	NA	NC
Median dose of steroid* at onset, mg/kg (range)	1 (0.0–5.0)	NA	NC
Median duration of steroid use, day (range)	48 (0–253)	NA	NC
Pre-engraft BSI, yes/no	10/33	39/234	.172
Acute GVHD			
Yes/no	36/7	168/105	.0047
Grades II–IV	35	112	
Grades III–IV	24	33	
Involved organ in acute GVHD			
Skin, yes/no	26/17	144/129	.345
GI tract, yes/no	31/12	66/207	<.0001
Liver, yes/no	5/38	17/256	.053
Initial treatment of acute GVHD	(n = 36)	(n = 168)	<.0001
None†	0	60	
mPSL < 1 mg/kg	3	11	
$1 \leq$ mPSL < 2 mg/kg	3	46	
$2 \text{ mg/kg} \leq$ mPSL	30	51	
Topically active BDP, yes/no	7/29	19/149	.184
Second-line therapies, yes/no	21/15	17/151	<.0001
mPSL pulse	5	2	
Etanercept	9	9	
MMF	4	3	
ATG	8	8	
MSC	2	1	

ANC indicates absolute neutrophil count; NA, not applicable; NC, not calculated; BDP, beclomethasone dipropionate; MMF, mycophenolate mofetil; MSC, mesenchymal stem cell.

* Methylprednisolone (mPSL) equivalent.

† Including topical steroids (cream, lotion, etc.).

cyclosporine A in patients who developed PE-BSI (15.4% [38/246] vs 7.1% [5/70]; $P = .074$).

We also found a significantly high incidence of PE-BSIs in patients with received RIC regimen (17.5% [32/183] versus 8.3% [11/133] with myeloablative conditioning regimen; $P = .018$) or in patients deemed as high-risk status at transplantation (16.9% [32/189] vs 8.7% [11/127] considered standard risk; $P = .035$). These data may reflect that patients who relapsed after HSCT preferentially received salvage transplantation from haploidentical donors by using RIC regimen including ATG, and GVHD prophylaxis consisted of tacrolimus plus mPSL at our institution, although a history of HSCT did not affect the incidence of PE-BSIs. There was no association between PE-BSI and age, sex, underlying disease, stem cell source, or HLA disparity (Table 1).

Clinical Findings in Patients with PE-BSI Patients

Clinical features of patients with PE-BSI are indicated in Table 3. The median onset of the first PE-BSI was 65–days (range, 18 to 154) after transplantation, and the median number of neutrophils at PE-BSI onset was $2.34 \times 10^9/L$ (range, .14 to 12.48). Four of 43 patients developed PE-BSI at the time of secondary neutropenia: 3 patients received ganciclovir for cytomegalovirus antigenemia, and another patient received histone deacetylase inhibitor for residual disease. Twenty-three of 43 patients (53.5%) with PE-BSI and 137 of 273 patients (50.2%) without PE-BSI experienced secondary neutropenia from the original engraftment until day +180, respectively, indicating that secondary neutropenia was not associated with an increased risk of PE-BSI ($P = .69$). Thirty-six of 43 patients (83.7%) who developed PE-BSI received corticosteroids, which were initiated at a median dose of 1–mg/kg mPSL equivalent, and treatment lasted 48–days (range, 0 to

253) before PE-BSI development. Ten of 43 PE-BSI patients developed BSI also in the pre-engraftment period, comparable with the 39 of 273 patients without PE-BSI who experienced pre-engraftment BSIs (23.3% vs 14.3%, $P = .172$). As consistent with previous reports [6–10], acute GVHD throughout the clinical course was significantly associated with PE-BSI. Acute GVHD (grades II to IV) occurred in 36 of 43 recipients (83.7%) who developed PE-BSI, whereas 112 of 273 patients (41.0%) did not develop PE-BSI ($P < .0001$). Moreover, severe acute GVHD (grades III to IV) was also a strong predictor for developing PE-BSI (55.8% vs 12.1%, $P < .0001$).

Notably, the incidence of PE-BSI was closely associated with the GI-GVHD. PE-BSI occurred more frequently in patients who developed GI-GVHD than in patients who did not develop GI-GVHD (32.0% [31/97] vs 5.5% [12/219], $P < .0001$), whereas no or a marginal relationship was observed between PE-BSI and skin or liver GVHD, respectively. To reflect the high proportion of GI involvement, an initial dose of ≥ 2 –mg/kg corticosteroids (mPSL equivalent) against acute GVHD was associated with PE-BSI development (37.0% vs 4.9%, $P < .0001$). In addition to corticosteroids, additional immunosuppressants (eg, etanercept, mycophenolate mofetil, and ATG) are usually administered at our institution, if acute GVHD is not sufficiently managed by corticosteroids. Expectedly, patients who required second-line treatment for acute GVHD had PE-BSIs develop more frequently than those who did not.

Subgroup Analysis Among Acute GI-GVHD Patients

Next, we performed the subgroup analyses focused on GI-GVHD patients ($n = 97$). We compared pre- and post-transplantation characteristics between 31 patients with PE-BSI and 66 patients without PE-BSI. As shown in Table 4, both the RIC regimen and mPSL use for GVHD prophylaxis were significantly

Table 4
Characteristics of the 97 Patients Who Developed GI-GVHD

	BSI		P
	Yes (n = 31)	No (n = 66)	
Median age, yr (range)	58 (23–69)	48.5 (16–68)	.054
Sex, male/female	21/10	33/33	.127
Underlying disease			.271
MDS/AML	13	29	
CML/MPN	0	5	
ALL	4	14	
Malignant lymphoma	13	15	
Plasma cell disorders	1	2	
AA	0	0	
Others	0	1	
Risk status at transplantation			.375
Standard/high	10/21	29/37	
Conditioning regimen			.025
Myeloablative/RIC	7/24	32/34	
Graft source			.266
BM/PB/CB	14/15/2	31/24/11	
Type of donor			.186
Matched related/partially matched related	4/10	13/11	
Matched unrelated/mismatched unrelated	8/9	16/26	
Prior transplantation			.505
No	21	49	
Autologous/allogeneic	4/6	2/15	
GVHD prophylaxis			
CsA/Tac	3/28	14/52	.252
sMTX, yes/no	16/15	46/20	.084
MMF, yes/no	4/27	13/53	.412
mPSL, yes/no	11/20	7/59	.003
ATG, yes/no	8/23	7/59	.054
PTCy, yes/no	0/31	3/63	.228
Pre-engraft BSI, yes/no	8/23	8/58	.184
Acute GVHD			
Grades II/III–IV	8/23	38/28	.004
Other involved organ in acute GVHD			
Skin, yes/no	21/10	44/22	1
Liver, yes/no	5/26	9/57	.231
Initial treatment of acute GVHD			.015
None*	0	5	
mPSL < 1 mg/kg	2	4	
1 ≤ mPSL < 2 mg/kg	3	21	
2 mg/kg ≤ mPSL	26	36	
Topically active BDP, yes/no	7/24	19/47	.957
Second-line therapies, yes/no	20/11	14/52	<.0001
mPSL pulse	5	1	
Etanercept	9	8	
MMF	4	3	
ATG	7	6	
MSC	2	0	
Other infectious complications			.188
None	12	31	
HHV-6 encephalomyelitis	3	2	
Hemorrhagic cystitis (ADV/BKV)	2/5	6/10	
Cytomegalovirus colitis	9	13	
Probable IA	0	3	
Others	7	12	

HHV-6 indicates human herpes virus 6; ADV/BKV, adenovirus/BK virus; IA, invasive aspergillosis.

* Including topical steroids (cream, lotion, etc.).

associated with PE-BSI. GVHD severity and its resistance to initial therapy also correlated with an increased risk of PE-BSI.

Finally, we assessed the correlation between PE-BSI and other infectious complications including cytomegalovirus gastroenteritis; however, we could not detect any significant difference.

Transplant Outcomes

Thirty-two of 43 patients with PE-BSI were dead on median 116 days (range, 26 to 726) after allo-HSCT; mortality was attributable to PE-BSI in 6 patients. Six patients died of uncontrolled PE-BSI, not disease relapse or exacerbation of GVHD

(n = 5) or without relapse in the absence of concomitant GVHD (n = 1). The median time from diagnosis of PE-BSI to death was 19.5 days (range, 9 to 37). In addition, PE-BSI contributed to the deaths of 9 patients with comorbidities, including GVHD, thrombotic microangiopathy, or multiple organ failure. Other causes of death were disease progression (n = 13) or GVHD with and without thrombotic microangiopathy (n = 4). Of the surviving patients, 2 relapsed and 9 remained disease-free. We found that probability of nonrelapse mortality at day 180 among patients with PE-BSI was significantly higher than that of patients without PE-BSI (37.8% vs 11.9%, $P < .001$). The median follow-up period was 36 months, and the 3-year OS after allo-HSCT was 18.6% for patients with PE-BSI, significantly lower than that of patients without PE-BSI (47.0%, $P < .001$) (Figure 1A). Our finding corroborates those of previous reports [9,12,13]. When we restricted our analysis to patients with acute GI-GVHD, the 3-year OS after allo-HSCT was 20.9% for patients with PE-BSI (n = 31), significantly lower than that for patients without PE-BSI (n = 66, 59.4%; $P < .001$) (Figure 1B).

Risk Factors for PE-BSI

Univariate analysis using logistic regression identified strong associations between the first PE-BSI episode and risk status at transplantation ($P = .035$), RIC regimen ($P = .021$), mPSL or ATG use for GVHD prophylaxis ($P < .0001$ and .027, respectively), and development of acute GVHD ($P = .007$), especially with GI tract involvement ($P < .0001$). Multivariate analysis confirmed that mPSL use for GVHD prophylaxis (odds ratio [OR], 6.49; 95% confidence interval [CI], 1.49 to 28.2; $P = .013$) and GI-GVHD (OR, 8.82; 95% CI, 3.99 to 19.5; $P < .0001$) were significantly associated with developing PE-BSIs. Among GI-GVHD patients, initiation of second-line therapies significantly increased the risk of PE-BSI (OR, 4.60; 95% CI, 1.46 to 14.5; $P = .009$) (Table 5).

DISCUSSION

We retrospectively evaluated the incidence, clinical features, and risk factors for PE-BSI among 316 allo-HSCT recipients. The overall incidence of PE-BSI in our study was 13.6%, in line with previous reports of 14% to 19.5% [6,8,15]. Several risk factors for PE-BSI have been described, including organ dysfunction (liver and kidney) [8], secondary neutropenia [8], HLA disparity [13], high-risk underlying disease [7], steroid use [7,11], and GVHD [6–12]. However, a study found no significant risk factors for PE-BSI [15]. In our study mPSL use for GVHD prophylaxis was identified as a risk factor for developing PE-BSI (OR, 6.49), as previously reported [23]. We mainly used prophylactic mPSL in the setting of ATG-combined haploidentical HSCTs for high-risk patients; however, mPSL use but not ATG use, disease status, or HLA disparity was statistically significant as determined via multivariate analysis. Therapeutic corticosteroid is also a well-known risk factor for infectious complications [7,11]. A recent study [24] revealed that the cumulative incidence rate of gram-negative rod bacteremia was 20% at 6 months after glucocorticoid treatment against severe acute GVHD; the median number of days from initiation of corticosteroids to the onset of gram-negative rod bacteremia was 55 (range, 10 to 579). We obtained a similar result in which 43 of 316 allo-HSCT recipients developed a PE-BSI at a median of 48–days after steroid treatment (Table 3). When patients without corticosteroid treatment were excluded, the median time from steroid initiation to PE-BSI development was 53.5–days (data not shown). Glucocorticoids can induce broad, nonspecific immunosuppressive and anti-inflammatory

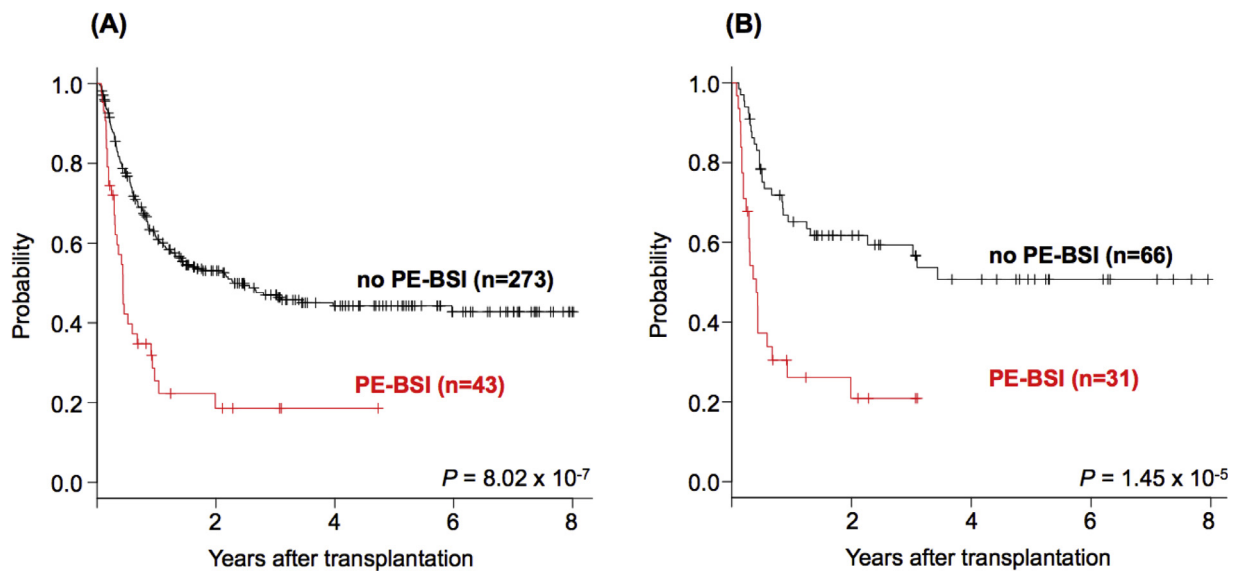


Figure 1. (A) Three-year OS after allo-HSCT in patients with or without PE-BSI (N = 316). (B) Three-year OS after allo-HSCT in 97 patients who had GI-GVHD either with or without PE-BSI.

Table 5
Results of Logistic Regression Analysis

Characteristics		Univariate OR (95% CI)	P	Multivariate OR (95% CI)	P
All patients					
Risk status at transplantation	Standard vs. high	0.52 (0.26–1.06)	.035	0.81 (0.35–1.89)	.629
Conditioning regimen	MAC vs. RIC	0.43 (0.21–0.88)	.021	0.57 (0.25–1.31)	.185
GVHD prophylaxis	mPSL usage	4.17 (2.04–8.52)	<.0001	6.49 (1.49–28.2)	.013
	ATG usage	2.32 (1.10–4.91)	.027	0.45 (0.10–2.05)	.302
Acute GVHD					
Skin	Yes vs. no	1.37 (0.71–2.64)	.347	0.93 (0.44–1.99)	.856
GI tract	Yes vs. no	8.10 (3.94–16.7)	<.0001	8.82 (3.99–19.5)	<.0001
Liver	Yes vs. no	1.98 (0.69–5.68)	.203	0.86 (0.26–2.89)	.805
Patients with GI-GVHD					
Severity of acute GVHD	Grade II vs. grades III–IV	3.90 (1.52–10.0)	.005	0.81 (0.23–2.78)	.732
Initial treatment of acute GVHD	2 mg/kg ≤ mPSL vs. others	4.33 (1.48–12.7)	.007	2.11 (0.61–7.36)	.240
Second-line therapies	Yes vs. no	6.75 (2.63–17.3)	<.0001	4.60 (1.46–14.5)	.009

MAC indicates myeloablative conditioning.

effects by suppressing lymphocyte proliferation and function by inhibiting IL-2 production. A lower number of lymphocytes on initiation of acute GVHD treatment correlated with subsequent gram-negative rod bacteremia as well as cytomegalovirus reactivation [24]. In addition, glucocorticoids, at high doses, suppress neutrophil functions by reducing the release of chemotactic factors from macrophages or the expression of adhesion molecules on endothelial cells [25]. Based on these findings, prophylactic mPSL might lead to lymphocytopenia and impaired neutrophil function, resulting in an increased risk of PE-BSI.

Another powerful predictor for PE-BSI was GI-GVHD (OR, 8.82); 31 of 97 patients (32.0%) with GI-GVHD developed PE-BSI. Intestinal stem cells and their niche Paneth cells reside within the crypts and play important roles in tissue remodeling and regeneration of the damaged epithelium. GI-GVHD targets these cells, disrupting the mucosal barrier and yielding various types of resident microbial pathogens to invade the bloodstream. A previous report [12] demonstrated that GI-GVHD predisposed to enteric bacterial BSIs; the incidence of enteric bacterial BSI was almost 3 times greater after GI-GVHD development than before in the first 120–days after allo-HSCT. In a mouse model, loss of Paneth cells led to

reduced secretion of α -defensins, antimicrobial peptides required for maintaining intestinal microbial ecology, resulting in a profound loss of diversity and an expansion of gram-negative bacteria in the intestinal microbiome [26–29]. Alterations in the intestinal microbiome have also been observed in allo-HSCT recipients with GI-GVHD [30]. In this study we found 62 causative pathogens of PE-BSI, among which 30 (*Enterococcus*, 8; gram-negative bacteria, 22) isolates were enteric bacteria [12] and 6 isolates were *Candida*, another member of intestinal commensal flora. The high proportion of enteric bacteria and *Candida* as causative pathogens, when compared with previous reports [6,13,15], may correspond to GVHD-induced intestinal dysbiosis.

There exists a vicious cycle between GI-GVHD and bacteremia; thus GI-GVHD is often resistant to standard corticosteroid treatments and requires additional intensive immunosuppressive treatment, which facilitates susceptibility for bacterial/fungal infection. Our subgroup analysis among patients with GI-GVHD showed that the additional requirement of immunosuppressants, rather than GVHD severity or initial dose of mPSL, was significantly associated with PE-BSI development, clearly indicating the importance for cutting off this vicious cycle to reduce the risk of subsequent PE-BSI. Another study [24] noted

that the addition of second-line treatment was associated with an increased risk of fungal diseases but not of gram-negative rod bacteremia.

Several reports have demonstrated a close relationship between PE-BSI and increased mortality [9,12,13], whereas other studies have not [6,8,15]. In this analysis we disclose that the development of PE-BSI was associated with inferior OS (18.6% vs 47.0%). Among 97 patients with GI-GVHD, development of PE-BSI was significantly associated with an adverse 3-year OS (20.9% vs 59.4%). GI-GVHD–induced or prophylactic/therapeutic antibiotics–induced intestinal dysbiosis might contribute to a higher mortality rate [30,31]; thus these aspects are viable therapeutic targets to improve patient outcomes of allo-HSCT. First, routine surveillance blood culture in patients with GI-GVHD, even if they are fever free, for early detection and prompt intervention of BSI is a recommended practice. Kameda et al. [32] revealed the benefit of surveillance blood culture to detect BSIs among allo-HSCT recipients treated with ≥ 0.5 –mg/kg corticosteroids. However, another report did not observe its utility [33]. In this study 27.9% of PE-BSI episodes were detected by surveillance blood culture, although it could not reduce the infection-related mortality (data not shown). We also highlighted the importance of routine surveillance culture, because causative pathogens of PE-BSI were detected during these procedures preceding blood culture tests in almost half of the patients with PE-BSI (41.9%). Such a routine practice may greatly assist in selecting antibiotics as empirical therapies.

Second, correcting intestinal dysbiosis is critically important, although only a report [34] demonstrated that intestinal dysbiosis was associated with subsequent bacteremia. Prophylactic administration of *Lactobacillus plantarum* (probiotics) has proven to be safe and feasible in children and adolescents undergoing allo-HSCT [35], whereas another randomized clinical trial using *Lactobacillus rhamnosus* GG failed to alter the intestinal microbiome and GVHD incidence [36]. In addition, several case reports have suggested that the probiotic itself caused septicemia in highly immunocompromised HSCT recipients [37–39]. Fecal microbiota transplantation may be an alternative treatment option for severe GI-GVHD. Fecal microbiota transplantation has been shown to affect microbiota composition with increased number of effector regulatory T cells in peripheral blood, thus ameliorating immune-mediated enteritis [40]. Finally, the use of prophylactic antibiotics in patients with steroid-refractory GI-GVHD should be carefully considered. Shono et al. [41] demonstrated that treatment of febrile neutropenia with broad-spectrum antibiotics (eg, imipenem-cilastatin or piperacillin-tazobactam) that possess robust activity against intestinal anaerobic commensal bacteria correlated with an increased risk of GVHD mortality both in allo-HSCT recipients and in the mouse model. Hence, it might be crucial to choose an antibiotic that spares anaerobic bacteria.

In conclusion, we identified mPSL use for GVHD prophylaxis and GI-GVHD as risk factors for PE-BSI after allo-HSCT. Our findings are limited given that this study was a single-center, retrospective analysis. An extremely high incidence of PE-BSI, preferentially because of enteric bacteria, in patients with steroid refractory GI-GVHD highlights the vicious cycle between GI-GVHD and bacterial infection. Improvement of acute GVHD treatment using novel agents, such as ruxolitinib, and appropriate supportive care for subsequent PE-BSI could reduce the risk of therapy-related mortality.

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